

Incremental backup

This is done by creating a full backup first, and then taking backups only for the data that changed since the last backup. This is different from a differential backup, where data that changed since the *last full backup* is copied.

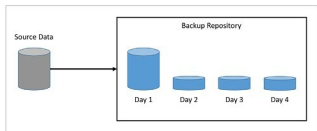


Figure 3: Incremental backup

For example, on Day 1 a full backup is created. On Day 2, only the data that changed since Day 1 is copied to the backup storage medium. On Day 3, only the changes in the source data since Day 2 are copied as backup data, and the same is done on Day 4. Incremental backup is better than differential backup because it takes less storage space and time to backup data. That's why this type of backup can also be done on optical discs like DVDs and Blu-ray.

Reverse-incremental backup

In a reverse-incremental backup, the incremental backups are injected into the full backup to create a synthetic full backup.

For example, on Day 1, a full backup is created. On Day 2, an incremental backup is taken for only the changed data, and is also injected into the full backup that was created on Day 1. The process is repeated on Day 3, Day 4, and so on, creating incremental and synthetic backups for each of these days. Though this backup process takes a little longer than a normal incremental backup, it takes very less time to restore the backup, as the synthetic full backup can be used for restoration easily. This type of backup can also be done on optical discs, but tapes are more suitable for creating the synthetic full backup.

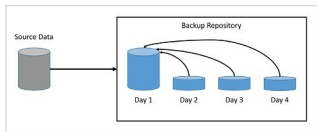


Figure 4: Reverse incremental backup

Among all these types of backups, the incremental and reverse-incremental backups are preferred. Such kinds of

backups can also be done for home and personal use, but one needs backup software to do that. Many backup software use a method called Continuous Data Protection (CDP), with the help of which the source data is automatically copied to the backup storage media continuously in real-time, whenever there is any change in it.

Storage media

Backup can be done on various storage media like hard drives, optical discs (CDs/DVDs) and tapes (digital audio tapes). A hard drive is the most common storage medium used for backups, but choosing the right hard drive and capacity is the biggest challenge. There are two types of hard drives — the Hard Disk Drive (HDD) and the Solid State Drive (SSD). HDDs are traditional hard drives, whereas SSDs are the latest ones and are much faster than the former. HDDs use a magnetic disk to store data and have mechanical parts, whereas SSDs use only microchips called NAND flash to store data and have no mechanical parts. That's why SSDs are more durable when compared to HDDs and are less prone to data loss from a physical impact/shock. However, SSDs are way too expensive when compared to HDDs and are not usually preferred for storing huge amounts of data.

While selecting a HDD one should look for the RPM of the drive. Drives with 7200 RPM are deemed better. The transfer rate of HDDs depends on their RPMs, and a SATA-III, 7200 RPM HDD can achieve a transfer rate of up to 300MB/s, whereas an SSD can reach up to a rate of 600MB/s. SSDs are faster than HDDs but can't hold data for a long period. A typical SSD can hold data for up to 10 years without power, but an HDD can hold data up to 30 years or more. SSDs use electrical charge to store data and because of internal electrical leakage, sometimes the data on these drives gets corrupted sooner than HDDs, whereas the latter use a magnetic disk to store data electro-magnetically. Choosing the capacity (GB) of the hard drive is also important to ensure enough storage is possible without wasting money on a drive with a high capacity.

Apart from the hard drive, backup can also be done using an optical disc. An optical disc is a kind of storage media on which data can be written or read using a laser beam. There are various types of optical discs like CDs, DVDs, and Blu-ray DVDs. A CD can store data up to 700MB, a DVD can store up to 4.7GB (single-sided, single-layer)/8.5GB (single-sided, double-layer) 9.4GB (double-sided, single-layer)/17.08 GB (double-sided, double-layered), and a Blu-ray has the capacity of 25GB (single-layer)/50GB (double-layer).

Creating backups on optical discs is also preferred because it's a cheaper option. There are two formats of optical discs — recordable (R) and rewritable (RW)/recordable erasable (RE). A recordable disc can be used only once for backup, whereas a rewritable disc can be used many times to